Reg. No. [ ] [ ] [ ] [ ] [ ] [ ] [ ] [ ] [ ] [ ] [MADIP41]

FOURTH SEMESTER B.E EXAMINATION, JUNE – 2016

DIPLOMA MATHEMATICS - II

Time: 3 Hours Maximum Marks:70

Instructions: Answer any **FIVE** full Questions.

1. a. Solve : $\frac{dy}{dx} = \sin(x+y) + \cos(x+y)$ 6

   b. Solve: $\frac{dy}{dx} = \frac{y+x-2}{y-x-4}$ 8

2. a. Solve : $(y^3 - 3x^2 y)dx - (x^3 - 3xy^2)dy = 0$ 4

   b. Solve: $\frac{dy}{dx} - \frac{2y}{x} = x + x^2$ 5

   c. Find the orthogonal trajectories of the family of parabolas $y^2 = 4ax$. 5

3. a. Solve $(x^2 - 3xy + 2y^2)\, dx + x\,(3x - 2y)\, dy = 0$ 4

   b. Solve $(x^2 + y^3 + 6x)\, dx + y^2 x\, dy = 0$ 5

   c. Solve $(1+xy)\, y\, dx + (1-xy)\, x\, dy = 0$ 5

4. a. Write a note on LCR circuits. 3

   b. Find the orthogonal trajectories of the family of astroids $x^{2/3} + y^{2/3} = a^{2/3}$ 6

   c. Is the family of polar curves $r^n = a \sin n\theta$, self orthogonal? 5

5. a. Solve: $(D^3 + 2D^2 + D)y = x^3$ 7

   b. Solve $\frac{dx}{dt} = 2x - 3y, \frac{dy}{dt} = y - 2x$ given x(0)=8 and y(0)=3 7

6. a. Express $\frac{5}{-3+4i}$ in the form of a + ib. 4

   b. Express $\frac{1}{\sin\alpha + i\cos\alpha}$ in the polar form. 5

   c. Find the modulus and amplitude of $\frac{2+i}{4i + (1+i)^2}$ 5

7. a. Find the analytic function whore real part is $\frac{\sin 2x}{\cosh 2y - \cos 2x}$ 7

   b. Find the bilinear transformation which maps the points 0, 1, ∞ into the points -5, -1, 3 respectively. 7

8. a. Verify Cauchy's theorem for the function $f(Z) = Z^2$, Where C is the square having vertices (0,0), (1,0), (1,1) and (0,1). 7

   b. Evaluate $\int_c \frac{e^Z}{Z - i\pi} dz$ where C is the circle

   i. $|z| = 2\pi$ ii. $|z| = \pi/2$